

E013

# Financial Planning for IT Professionals

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

An IT department buys \$120,000 of servers, delivered on a Tuesday, that are expected to last five years. It also signs up for a cloud monitoring service at \$2,000 a month from a vendor based in Canada. Which spending is a capital expenditure?

- A. Neither, because all IT spending is operating expense.
- B. The monitoring subscription.
- C. Both, since both support IT operations.
- D. The server purchase.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

A company stops buying its own servers and moves to a pay-as-you-go cloud service. How does this change its IT spending mix?

- A. It shifts spending from CapEx toward OpEx.
- B. It shifts spending from OpEx toward CapEx.
- C. It eliminates IT costs entirely.
- D. It leaves the mix unchanged.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A help-desk automation tool, recommended by a peer at a conference, costs \$80,000. It will save \$30,000 a year for four years. The help desk has nine agents. What is the ROI over the four years?

- A. About 33%
- B. 50%
- C. 37.5%
- D. 150%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**4.** [TRANSFER — reversed stem / different variable]

A project costs \$50,000. Leadership requires a 40% ROI. What total benefit must the project deliver?

- A.** \$50,000
- B.** \$20,000
- C.** \$70,000
- D.** \$125,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**5.**

A new storage array rated for 99.999% uptime costs \$90,000. It will cut maintenance costs by \$25,000 a year and power costs by \$5,000 a year. What is the payback period?

- A.** 3.6 years
- B.** 3 years
- C.** About 0.33 years
- D.** 18 years

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**6.** [TRANSFER — reversed stem / different variable]

A \$60,000 project must pay for itself within 2 years. What annual savings does it need?

- A.** \$60,000 a year
- B.** \$30,000 a year
- C.** \$120,000 a year
- D.** Any positive amount, since every project eventually pays back.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**7.**

A project proposed by the infrastructure team's most senior engineer costs \$10,000 today and will return \$11,000 one year from now. The company's discount rate is 12%. What is the NPV, and should the project proceed?

- A.** +\$1,000; accept it.
- B.** +\$2,320; accept it.
- C.** About -\$179; reject it.
- D.** Accept it, since a senior engineer proposed it.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**8.** [TRANSFER — reversed stem / different variable]

An investment of \$10,000 today returns \$11,000 in one year. At what discount rate is its NPV exactly zero?

- A.** 10%
- B.** 0%, because NPV is zero only when there is no return.
- C.** About 9.1%
- D.** 110%

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**9.**

A company plans to buy 100 laptops at \$1,200 each, with support at \$150 per laptop per year. The laptops come in two colors, and the vendor is headquartered in Texas. What is the three-year total cost of ownership for the fleet?

- A.** \$1,650
- B.** \$135,000
- C.** \$120,000
- D.** \$165,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**10.**

Network equipment installed across three branch offices cost \$60,000. It has a five-year useful life and a \$6,000 salvage value. Using straight-line depreciation, what is the annual depreciation expense?

- A.** \$10,800
- B.** \$54,000
- C.** \$13,200
- D.** \$12,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A \$200,000 annual software license is shared by three departments and charged back by number of users: Sales has 250 users, Operations 150, and HR 100. Sales generates 60% of company revenue. How much is charged to Sales?

- A.** About \$66,667
- B.** \$40,000
- C.** \$100,000
- D.** \$120,000

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

**12.**

A custom inventory system, the former CIO's signature initiative, has consumed \$500,000. Finishing it will cost \$300,000 more. An off-the-shelf product with the same capability costs \$150,000. What should the company do?

- A.** Finish it to honor the former CIO's initiative.
- B.** Finish it, since abandoning a project always wastes money.
- C.** Finish the custom system because \$500,000 is already invested.
- D.** Switch to the off-the-shelf product.

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

An IT department buys \$120,000 of servers, ~~delivered on a Tuesday~~, that are expected to last five years. It also signs up for a cloud monitoring service at \$2,000 a month ~~from a vendor based in Canada~~. Which spending is a capital expenditure?

**Target word:** classify CapEx **Decides:** The servers are long-lived assets.

|   |                   |                                                      |
|---|-------------------|------------------------------------------------------|
| A | Absolute language | Hardware purchases are commonly capitalized.         |
| B | Confusion set     | A recurring service fee is an operating expense.     |
| C | Dropped element   | Drops the asset-vs-recurring distinction.            |
| D | <b>Correct</b>    | Long-lived equipment is capitalized and depreciated. |

## 2. Answer A Concept 1 • transfer

A company stops buying its own servers and moves to a pay-as-you-go cloud service. How does this change its IT spending mix?

**Target word:** direction of the shift **Decides:** Owned assets become recurring fees.

|   |                    |                                                  |
|---|--------------------|--------------------------------------------------|
| A | <b>Correct</b>     | Subscriptions replace capital purchases.         |
| B | Inverted operation | Reverses the direction.                          |
| C | Absolute language  | Cloud services still cost money.                 |
| D | Dropped element    | Drops the change in how the service is paid for. |

## 3. Answer B Concept 2

A help-desk automation tool, ~~recommended by a peer at a conference~~, costs \$80,000. It will save \$30,000 a year for four years. ~~The help desk has nine agents~~. What is the ROI over the four years?

**Target word:** four-year ROI **Decides:** Total benefit =  $4 \times \$30,000 = \$120,000$ .

|   |                    |                                                       |
|---|--------------------|-------------------------------------------------------|
| A | Inverted operation | Divides the gain by the benefit instead of the cost.  |
| B | <b>Correct</b>     | $(\$120,000 - \$80,000) / \$80,000 = 50\%$ .          |
| C | Wrong stem number  | Uses one year of savings ( $\$30,000 / \$80,000$ ).   |
| D | Dropped element    | $\$120,000 / \$80,000$ drops the subtraction of cost. |

## 4. Answer C Concept 2 • transfer

A project costs \$50,000. Leadership requires a 40% ROI. What total benefit must the project deliver?

**Target word:** solve for benefit **Decides:** Benefit = cost  $\times (1 + \text{ROI})$ .

|   |                    |                                                      |
|---|--------------------|------------------------------------------------------|
| A | Wrong stem number  | Recovering only the cost gives 0% ROI.               |
| B | Dropped element    | That is only the gain; it drops the cost recovery.   |
| C | <b>Correct</b>     | $\$50,000 \times 1.40 = \$70,000$ .                  |
| D | Inverted operation | Divides cost by 0.40 instead of multiplying by 1.40. |

### 5. Answer B Concept 3

A new storage array ~~rated for 99.999% uptime~~ costs \$90,000. It will cut maintenance costs by \$25,000 a year and power costs by \$5,000 a year. What is the payback period?

**Target word:** payback period **Decides:** Annual savings = \$25,000 + \$5,000 = \$30,000.

|          |                    |                                    |
|----------|--------------------|------------------------------------|
| A        | Dropped element    | Uses only the maintenance savings. |
| <b>B</b> | <b>Correct</b>     | $\$90,000 / \$30,000 = 3$ .        |
| C        | Inverted operation | Divides savings by cost.           |
| D        | Wrong stem number  | Uses only the power savings.       |

### 6. Answer B Concept 3 • transfer

A \$60,000 project must pay for itself within 2 years. What annual savings does it need?

**Target word:** solve for annual savings **Decides:** Savings = cost / payback period.

|          |                    |                                           |
|----------|--------------------|-------------------------------------------|
| A        | Wrong stem number  | That would pay back in one year.          |
| <b>B</b> | <b>Correct</b>     | $\$60,000 / 2 = \$30,000$ .               |
| C        | Inverted operation | Multiplies cost by the period.            |
| D        | Absolute language  | The 2-year limit sets a specific minimum. |

### 7. Answer C Concept 4

A project ~~proposed by the infrastructure team's most senior engineer~~ costs \$10,000 today and will return \$11,000 one year from now. The company's discount rate is 12%. What is the NPV, and should the project proceed?

**Target word:** NPV and decision **Decides:** PV of \$11,000 at 12% = \$9,821.

|          |                    |                                                                           |
|----------|--------------------|---------------------------------------------------------------------------|
| A        | Dropped element    | Drops discounting and compares dollars from different years.              |
| B        | Inverted operation | Multiplies the future cash by 1.12 instead of dividing.                   |
| <b>C</b> | <b>Correct</b>     | $\$11,000 / 1.12 \approx \$9,821$ ; $\$9,821 - \$10,000 \approx -\$179$ . |
| D        | Camouflage         | Who proposed it does not change the math.                                 |

### 8. Answer A Concept 4 • transfer

An investment of \$10,000 today returns \$11,000 in one year. At what discount rate is its NPV exactly zero?

**Target word:** solve for the rate **Decides:**  $\$10,000 \times (1 + r) = \$11,000$ .

|          |                    |                                                                  |
|----------|--------------------|------------------------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | $\$11,000 / \$10,000 - 1 = 10\%$ .                               |
| B        | Absolute language  | NPV is zero when the return equals the discount rate.            |
| C        | Confusion set      | Divides the gain by the future amount instead of the investment. |
| D        | Inverted operation | Reports the ratio, not the rate of return.                       |

## 9. Answer D Concept 5

A company plans to buy 100 laptops at \$1,200 each, with support at \$150 per laptop per year. ~~The laptops come in two colors, and the vendor is headquartered in Texas.~~ What is the three-year total cost of ownership for the fleet?

**Target word:** three-year fleet TCO **Decides:** Purchase  $\$120,000$  + support  $100 \times \$150 \times 3 = \$45,000$ .

|   |                   |                                          |
|---|-------------------|------------------------------------------|
| A | Adjacent category | The per-laptop TCO, not the fleet total. |
| B | Wrong stem number | Counts only one year of support.         |
| C | Dropped element   | Leaves out support.                      |
| D | <b>Correct</b>    | $\$120,000 + \$45,000 = \$165,000$ .     |

## 10. Answer A Concept 6

Network equipment ~~installed across three branch offices~~ cost \$60,000. It has a five-year useful life and a \$6,000 salvage value. Using straight-line depreciation, what is the annual depreciation expense?

**Target word:** annual depreciation **Decides:**  $(\text{Cost} - \text{salvage}) / \text{life}$ .

|   |                    |                                                       |
|---|--------------------|-------------------------------------------------------|
| A | <b>Correct</b>     | $(\$60,000 - \$6,000) / 5 = \$10,800$ .               |
| B | Adjacent category  | The total depreciable amount, not the annual expense. |
| C | Inverted operation | Adds the salvage value instead of subtracting it.     |
| D | Dropped element    | Ignores the salvage value.                            |

## 11. Answer C Concept 7

A \$200,000 annual software license is shared by three departments and charged back by number of users: Sales has 250 users, Operations 150, and HR 100. ~~Sales generates 60% of company revenue.~~ How much is charged to Sales?

**Target word:** Sales' allocation **Decides:** Sales has 250 of 500 users (50%).

|   |                   |                                                    |
|---|-------------------|----------------------------------------------------|
| A | Confusion set     | Splits evenly, ignoring the driver.                |
| B | Adjacent category | That is HR's allocation.                           |
| C | <b>Correct</b>    | $250 / 500 \times \$200,000 = \$100,000$ .         |
| D | Camouflage        | Uses the revenue share instead of the user driver. |

## 12. Answer D Concept 8

A custom inventory system, ~~the former CIO's signature initiative~~, has consumed \$500,000. Finishing it will cost \$300,000 more. An off-the-shelf product with the same capability costs \$150,000. What should the company do?

**Target word:** decision ignoring sunk cost **Decides:** Future cost: \$300,000 to finish vs. \$150,000 to switch.

|   |                   |                                                                                          |
|---|-------------------|------------------------------------------------------------------------------------------|
| A | Camouflage        | History and sponsorship do not change future costs.                                      |
| B | Absolute language | Stopping can be the cheaper path forward.                                                |
| C | Confusion set     | The sunk-cost fallacy: past spending cannot be recovered.                                |
| D | <b>Correct</b>    | The \$500,000 is sunk; going forward, \$150,000 beats \$300,000 for the same capability. |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 8             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 5             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 3             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 8             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 5             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 4             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 3             |             |

## Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                            | Rule                                                                                                                                                                 | Trap                                                                        | Tell                                                     |
|---|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | <b>CapEx vs. OpEx</b>              | Capital expenditures buy long-lived assets that are depreciated; operating expenditures are recurring costs expensed as incurred, such as subscriptions and support. | Classifying by dollar size or by whether it is “IT.”                        | A purchase of equipment next to a monthly or annual fee. |
| 2 | <b>Return on investment</b>        | $ROI = (\text{total benefit} - \text{cost}) / \text{cost}$ .                                                                                                         | Dividing benefit by cost without subtracting, or using one year of benefit. | A cost and a stream of savings over several years.       |
| 3 | <b>Payback period</b>              | Payback = upfront cost / annual net savings, counting every recurring saving.                                                                                        | Leaving out one of the savings streams or inverting the ratio.              | “How long until it pays for itself?”                     |
| 4 | <b>Time value of money and NPV</b> | Discount future cash to today’s dollars; accept a project only if NPV (present value of inflows – cost) is positive.                                                 | Comparing future dollars to today’s dollars at face value.                  | A discount rate and cash arriving in a later year.       |
| 5 | <b>Total cost of ownership</b>     | TCO includes purchase price plus every recurring cost over the full life of the asset.                                                                               | Stopping at the sticker price or counting one year of support.              | Unit price plus per-year support over several years.     |
| 6 | <b>Straight-line depreciation</b>  | Annual depreciation = $(\text{cost} - \text{salvage value}) / \text{useful life}$ .                                                                                  | Ignoring or adding the salvage value.                                       | An asset cost, a salvage value, and a life in years.     |
| 7 | <b>Chargeback by cost driver</b>   | Allocate shared IT costs by the driver that causes them (users, devices, usage), in proportion to each department’s share.                                           | Splitting evenly or by an unrelated measure like revenue.                   | A shared cost and per-department driver counts.          |
| 8 | <b>Sunk costs</b>                  | Money already spent cannot be recovered and should not affect future choices; compare only future costs and benefits.                                                | Continuing a project because so much has been spent.                        | “Already invested” next to a cheaper alternative.        |

**Course scope (public description):** Financial Planning for Information Technology (IT) Professionals provides a foundational basis for technology leaders to determine the value and cost of an organization’s technology. Through this course, students will learn the essentials of financial literacy, as it relates to their role in the technology branch of an organization. Students will read and interpret key indicators such as budgets and tracking tools, benchmark the value of technologies within an organization, and predict the return on investment for a technological change. These are highly valuable skills to determine practical solutions to business problems, while reducing unnecessary costs. There are no prerequisites for this course.